

FORBES, Australia

WMO: 947150

Lat: 33.3628S

Lon: 147.9206E

Elev: 231

StdP: 98.58

Time Zone: 10.00 (AUS)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-1.0	0.2	-6.6	2.2	22.5	-4.5	2.7	17.9	10.3	13.9	9.4	13.5	0.7	180	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.8	38.7	19.1	36.6	18.8	34.8	18.5	22.3	28.9	21.6	28.8	20.8	28.8	5.3	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	16.2	23.2	19.9	15.0	23.1	18.6	13.8	22.8	66.7	28.5	63.9	28.5	61.2	28.8	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.9	7.9	-3.3	41.7	1.4	2.0	-4.3	43.2	-5.1	44.4	-5.9	45.5	-6.9	47.0		
(5)			-3.4	23.7	1.4	1.0	-4.4	24.5	-5.2	25.1	-6.0	25.7	-7.0	26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.8	26.2	25.0	21.7	17.0	12.3	9.5	8.6	9.4	12.5	16.5	20.4	23.2		
	DBStd	6.92	3.64	3.52	3.35	3.25	2.91	2.53	2.31	2.46	3.07	3.75	4.03	3.67		
	HDD10.0	156	0	0	0	1	11	40	56	41	8	1	0	0		
	HDD18.3	1388	0	1	9	61	188	266	302	276	176	83	22	4		
	CDD10.0	2643	503	420	362	210	82	24	12	23	84	202	312	409		
	CDD18.3	833	245	187	113	21	1	0	0	0	2	25	84	156		
	CDH23.3	10717	3228	2233	1189	280	10	0	0	1	64	446	1203	2063		
(13)	CDH26.7	5338	1844	1176	487	66	0	0	0	0	10	154	561	1039		
(14)	Wind	WSAvg	3.5	4.3	4.0	3.6	2.9	2.8	2.9	3.2	3.5	3.7	4.1	4.1		
(15) Precipitation	PrecAvg	558	61	42	40	40	54	33	47	47	47	55	48	45		
	PrecMax	934	231	171	128	299	156	79	144	106	147	180	176	136		
	PrecMin	236	0	0	0	0	6	1	4	1	9	5	2	0		
	PrecStd	162	56	44	33	54	38	21	30	27	35	38	40	38		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.0	40.5	35.8	31.6	24.6	20.1	19.2	22.3	28.0	33.2	38.6	39.5		
		MCWB	19.9	20.5	18.1	16.9	15.1	13.1	12.1	12.7	14.5	15.8	17.8	18.5		
	2%	DB	39.2	37.6	33.4	28.5	22.6	17.9	17.0	19.5	25.0	30.6	35.2	36.5		
		MCWB	19.3	19.5	18.0	16.4	14.3	12.6	11.4	12.1	13.9	15.7	17.0	18.2		
	5%	DB	37.1	35.2	31.3	26.5	20.8	16.3	15.4	17.7	22.8	28.1	32.3	34.4		
		MCWB	19.4	19.2	17.8	15.6	13.6	11.9	11.1	11.2	13.5	15.2	16.6	17.7		
	10%	DB	35.0	33.2	29.4	24.6	18.8	14.8	13.9	15.9	20.5	25.6	29.8	32.3		
		MCWB	19.1	18.7	17.4	15.0	12.8	11.6	10.3	10.9	12.9	14.8	15.9	17.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.1	23.8	22.5	19.6	17.4	15.3	15.1	15.1	17.6	19.3	20.8	22.5		
		MCDB	31.1	27.9	26.1	24.3	20.0	16.5	16.1	17.9	22.1	24.7	27.2	27.8		
	2%	WB	22.1	22.5	21.2	18.1	15.8	13.9	12.9	13.4	16.1	18.0	19.6	21.1		
		MCDB	30.6	28.7	25.8	23.5	20.2	15.7	15.2	17.1	21.0	24.1	27.2	27.5		
	5%	WB	21.4	21.6	20.1	17.1	14.7	12.9	11.7	12.3	15.0	17.1	18.7	20.1		
		MCDB	30.7	28.5	25.9	23.1	18.7	15.4	14.1	16.4	20.1	23.8	26.2	27.4		
	10%	WB	20.7	20.8	19.1	16.2	13.7	11.9	10.7	11.1	13.8	16.1	17.7	19.1		
		MCDB	30.5	28.6	25.9	22.6	17.8	14.4	13.2	15.2	18.5	23.1	25.5	27.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.8	13.8	13.5	14.1	12.9	10.7	10.8	12.6	14.4	15.2	15.2	15.2		
		MCDBR	17.2	16.5	16.1	16.6	15.3	13.0	13.7	15.9	18.4	19.0	18.8	17.7		
		MCWBR	5.0	5.3	5.8	7.1	8.0	8.1	8.4	9.0	8.9	7.5	6.2	5.7		
		MCDBR	13.6	12.0	11.7	12.9	11.9	10.4	10.5	12.9	13.8	14.3	13.6	13.3		
	5% WB	MCWBR	5.1	4.8	5.3	6.1	6.8	7.0	7.2	8.1	7.8	6.7	5.2	5.4		
(40) Clear-Sky Solar Irradiance	taub		0.346	0.347	0.323	0.306	0.291	0.285	0.281	0.288	0.318	0.328	0.343	0.338		
	taud		2.498	2.523	2.568	2.584	2.585	2.608	2.610	2.567	2.467	2.468	2.470	2.508		
	Ebn at Noon		991	970	959	924	887	867	890	929	948	978	987	1003		
	Edh at Noon		114	108	96	84	74	68	71	84	104	112	117	114		
(44) All-Sky Solar Radiation	RadAvg		7.70	6.71	5.70	4.45	3.25	2.50	2.80	3.73	5.17	6.40	7.08	7.77		
	RadStd		0.38	0.48	0.37	0.29	0.20	0.20	0.18	0.28	0.38	0.47	0.62	0.31		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	+0.47	N/A	N/A	+1.28	N/A	N/A	N/A	-88	+191	+112

Nomenclature: See separate page